MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. If the angle made by the rope with the ground level is 30°, the height of the pole is: [1]

(A) 10 m (B) 20 m (C) $10\sqrt{3}$ m (D) 5 m

Q2. If the angles of elevation of the top of a tower from two points on the ground at distances a and b from the base are 30° and 60° respectively, then the height of the tower is: [1]

(A) $\sqrt{a+b}$ (B) $\sqrt{ab}$ (C) $\dfrac{a+b}{2}$ (D) ab

Q3. A tree breaks due to a storm and the broken part bends so that the top of the tree touches the ground making an angle of 30° with it. The distance from the foot of the tree to the point where the top touches the ground is 8 m. The total height of the tree is: [1]

(A) $8\sqrt{3}$ m (B) $\dfrac{8\sqrt{3}}{3}$ m (C) $16\sqrt{3}$ m (D) 8 m

Q4. The angle of depression of a point on the ground from the top of a 60 m high building is 45°. The distance of the point from the foot of the building is: [1]

(A) 30 m (B) 60 m (C) $60\sqrt{3}$ m (D) $30\sqrt{2}$ m

Q5. **Assertion (A):** A man standing on a platform 3 m above the surface of a lake observes a cloud and its reflection in the lake. The angle of elevation of the cloud is always strictly less than the angle of depression of its reflection.

Reason (R): The reflection is located at a depth below the water surface equal to the height of the cloud above the water surface, making the vertical distance from the observer to the reflection greater than that to the cloud. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

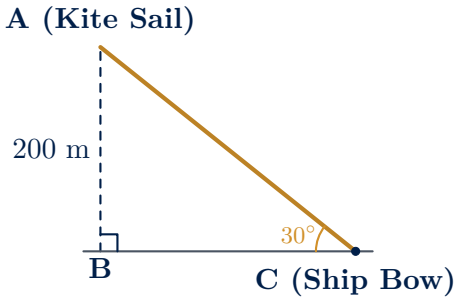
Q6. A 6 m high vertical pole casts a shadow $2\sqrt{3}$ m long on the ground. At the same time, a nearby tower casts a shadow 30 m long. Find the height of the tower. [2]

Q7. From the top of a 60 m high building, the angles of depression of the top and bottom of a transmission tower are observed to be 30° and 60° respectively. Find the height of the transmission tower. [3]

Q8. A man on top of a vertical tower 150 m high spots two cars parked on the same straight road leading to the tower. If their angles of depression are 30° and 60° respectively, find the distance between the two cars. [3]

Q9. A boy of height 1.3 m spots a balloon moving with the wind in a horizontal level at some height from the ground. The angle of elevation of the balloon from his eyes at any instant is 60°. After 2 seconds, the angle of elevation reduces to 30°. If the horizontal speed of the wind is $29\sqrt{3}$ m/s, find the height of the balloon from the ground. [3]

Q10. CASE STUDY: Skysails Green Maritime Propulsion



Modern ocean cargo vessels employ auxiliary wind propulsion systems known as 'Skysails' to pull ships through open sea water. A massive computer-controlled towing kite flies at altitudes between 100 m and 300 m to capture high-altitude winds, significantly reducing vessel fuel consumption.

A commercial cargo vessel deploys a kite sail that ascends to an operating vertical height of 200 m above sea level ($AB = 200$ m). The towing cable is anchored to the vessel's bow at point C. At a particular wind speed, the tow cable makes an angle of inclination $\theta = 30°$ with the horizontal.

Based on the above mathematical scenario, answer the following questions:

(i) Find the length of the tow cable paid out (AC) when the kite operates at an inclination of 30° to the horizontal. **[1]**

(ii) Find the horizontal distance between the vessel's bow (C) and point B directly beneath the kite. **[1]**

(iii) If changing wind patterns cause the cable inclination to increase to 45° while maintaining the same vertical height of 200 m, find the new length of the tow cable paid out. [Take $\sqrt{2} = 1.414$] **[2]**

— **OR** —

Find the decrease in horizontal distance between the vessel's bow and point B when the inclination increases from 30° to 45°. **[2]**

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★★★ **END OF QUESTION PAPER · ALL THE BEST** ★★★